REZON fall detector · parts list

Hackathon build with the SM-24 geophone and a SIM800L call alert · casing 3D-printed on site · no breadboard

2026-11-10
Row = line in the Hestore stock CSV

SIGNAL CHAIN

SM-24 geophone → ADC 15 Click → SPI → ESP32-S3

ESP32-S3 → I²S → PCM5102A → PAM8403 → Speaker

ESP32-S3 → UART → SIM800L → GSM call → our phone

Plus a WS2812B status LED (green / amber / red) and an "I'm OK" button. One 18650 cell feeds the SIM800L directly and a 5 V boost for everything else.

OURS (NOT IN HACKATHON STOCK)

- **SM-24 geophone** (arriving today)
- **MikroE ADC 15 Click** (ADS131M02, already running on the bench)
- **SIM800L GSM/GPRS module** with its antenna
- **SIM card from one of our phones** + nano → micro SIM adapter
- **TP4056 USB-C charger module** with DW01 protection

SIM800L SETUP

- Power it **straight from the 18650** (3.4–4.4 V). Never from the 5 V boost: 5 V destroys it.
- Bursts up to 2 A while transmitting: **1000 µF + 100 nF** right at its VCC/GND pins, short thick wires.
- Logic is 2.8 V: ESP32 TX → SIM RX through a **10 kΩ / 20 kΩ divider**. SIM TX → ESP32 RX direct.
- ESP32-S3 UART1 on free GPIOs (not 33–37), plus one GPIO to the SIM800L RST pin.
- SIM800L is **2G only**: check the SIM's network still has 2G coverage at the venue.
- Turn the **SIM PIN off** in the phone first; test the call with AT commands before the show.

Sensing & processing

	QTY	PART	ROW	SOURCE	PURPOSE
☐	1	SM-24 geophone (SparkFun SEN-11744)	–	Ours	Floor vibration sensor, 28.8 V/(m/s), 10–240 Hz
☐	1	MikroE ADC 15 Click (ADS131M02, 24-bit, 2 ch) `MIKROE-4890`	–	Ours	Geophone straight into the ADC at gain 32 (±37.5 mV full scale); firmware already brought up
☐	1	ESP32 S3 fejlesztői modul, 16MB Flash, 8MB PSRAM, WiFi, BT5, USB-C `ESP32-S3-WROOM-1-N16R8-M`	179	Hackathon	Main controller; exact target of our firmware (PSRAM for buffers and the model)
☐	1	16 bites analog-digital konverter modul, I2C BACKUP `ADS1115-M`	11	Hackathon	Only if the Click fails: ±0.256 V range, max 860 SPS; enough for fall impacts

Alert: call, voice, light, button

	QTY	PART	ROW	SOURCE	PURPOSE
☐	1	SIM800L GSM/GPRS modul, with spring or IPEX antenna	–	Ours	Calls our phone when a fall is confirmed and nobody presses "I'm OK"
☐	1	SIM card from one of our phones (PIN disabled)	–	Ours	Network access for the call; must work on 2G
☐	1	Nano → micro SIM adapter	–	Ours	The SIM800L takes a micro SIM; phones use nano
☐	1	PCM5102A alapú DAC audió dekóder I2S modul	249	Hackathon	Audio out for "Mari néni, jól van?"; the S3 has no built-in DAC
☐	1	Hangszóró erősítő, PAM8403 alapú 2x3W erősítő modul	269	Hackathon	Speaker amplifier, runs from the 5 V rail
☐	1	Hangszóró, kicsi 1W 8ohm	262	Hackathon	Speaker
☐	1	WS2812B RGB vezérelhető LED szalag, 1 m, 60 LED/m `WS2812B-LS-60-IP20-BK`	68	Hackathon	Cut one pixel: green / amber / red status LED
☐	1	1N4001 dióda	364	Hackathon	Drops the LED supply to ~4.3 V so it reliably reads the 3.3 V data line
☐	2	TACT mikrokapcsoló, SPST-NO, THT, 6x6mm, 5mm `TACT-65R-F`	51	Hackathon	"I'm OK" / cancel-alarm button (+1 spare)

Power

	QTY	PART	ROW	SOURCE	PURPOSE
☐	**2**	Li-Ion akkumulátor, 18650, 3.6V, 3500mAh `INR18650-MJ1`	2	**Hackathon**	Battery (+1 spare); handles the SIM800L's 2 A bursts easily
☐	**1**	Elem / akkumulátortartó, 1 x 18650, vezetékkel `1 X 18650 W`	3	**Hackathon**	Battery holder
☐	**1**	Világítós billenőkapcsoló, 12V, SPST, OFF-ON `AK-01`	46	**Hackathon**	Main power switch; carries the 2 A bursts (the small slide switch is only rated 0.5 A)
☐	**1**	DC-DC boost	304	**Hackathon**	Cell → 5 V rail for the ESP32 5V pin, amp, DAC and LED only (not the SIM800L); set to 5.0 V
☐	**1**	TP4056 USB-C charger module with DW01 protection	–	**Ours**	Battery charging; no charger on the hackathon list
☐	**1**	Fémréteg ellenállás készlet, 500 db, 50 érték, 0.5W, 1% `RESISTOR-KIT-500P`	165	**Hackathon**	1 kΩ across the geophone coil · 10 k / 20 k SIM800L RX divider · 2 × 100 kΩ battery divider · 10 kΩ button pull-up
☐	**1**	Elektrolit kondenzátor készlet, 0.1uF–1000uF, 500 db `ELCOPACK-500`	166	**Hackathon**	1000 µF at the SIM800L VCC · 470–1000 µF on the 5 V rail
☐	**1**	Kerámia kondenzátor készlet, 24 féle, 960 db `CER-CAP-PACK`	167	**Hackathon**	100 nF decoupling at each module's supply, including the SIM800L

Wiring & assembly

	QTY	PART	ROW	SOURCE	PURPOSE
☐	**1**	9x15 cm, kétoldalas furatgalvánozott, univerzális nyák, FR4 `DPCB-915`	101	**Hackathon**	Base board; room for the SIM800L next to the other modules (cut to fit the case)
☐	**2**	Tüskesor aljzat, egyenes, 40 pin `ZL262-40SG`	131	**Hackathon**	Sockets so the ESP32-S3, Click and SIM800L stay swappable
☐	**1**	Tüskesor, egyenes, tördelhető, 40 pin `ZL201-40G (WWS40G)`	130	**Hackathon**	Pin headers for modules that lack them
☐	**1**	Szalagkábel csatlakozóval, 10cm, 40p, female-female `RC-40-10/FF`	42	**Hackathon**	Click mikroBUS → ESP32-S3 (SCK, SDO, SDI, CS1, CS2, INT, RST, 3V3, GND)
☐	**1**	kábel 6x0,22	359	**Hackathon**	Geophone lead (one twisted pair), about 1 m
☐	**1**	Vezeték, H05V-K, sodrat, Cu, 1 x 0.5mm², piros	353	**Hackathon**	Battery + / 5 V / SIM800L VCC
☐	**1**	Vezeték, H05V-K, sodrat, Cu, 1 x 0.5mm², fekete	352	**Hackathon**	GND
☐	**1**	Egymagos vörösréz huzal 20 AWG, 15 m	349	**Hackathon**	Links on the perfboard
☐	**1**	Zsugorcső készlet (2:1), 530 db `HST-SET530`	157	**Hackathon**	Insulation
☐	**1**	USB 2.0 kábel, USB A – USB C, 3A, 1.5m `CTHBG`	138	**Hackathon**	Flashing and serial monitor
☐	**1**	M3 távtartó készlet	205	**Hackathon**	Mount the board in the printed case
☐	**1**	Csavar készlet M2–M6	302	**Hackathon**	Fixings
☐	**1**	Kábelkötegelő (100 db), fekete, 2.5 × 100 mm `CV 100/2,5/B`	64	**Hackathon**	Cable management and strain relief

Second SM-24, if it turns up: same wiring on the Click's CH2, also with a 1 kΩ resistor across the coil.

Left out on purpose: casing, breadboards, soldering tools, displays, other accelerometers.